Unfolding with TRExFitter

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Top Reconstruction Meeting - 20/02/2020

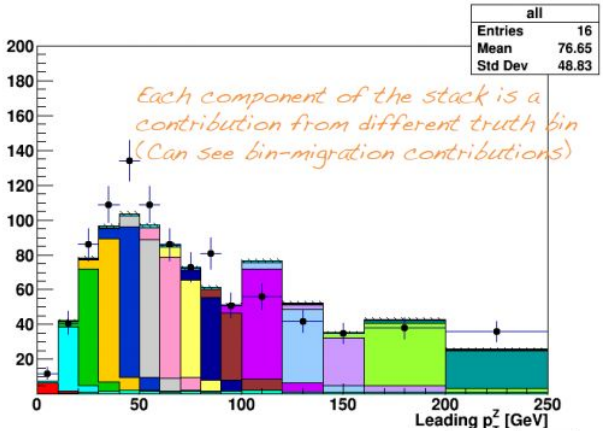
Profile-likelihood unfolding (PLU)

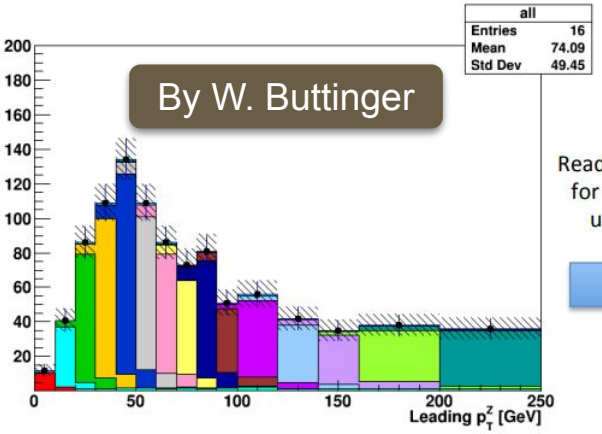
- **The idea:**
  - measure single- and multi-dimensional differential cross sections using a profile likelihood fit
  - see these slides by Will Buttinger:
    https://indico.cern.ch/event/670305/contributions/2742204/attachments/1535980/2406195/troofitUnfolding.pdf
  - method, plans and first tests presented already in TopXS meetings:
    https://indico.cern.ch/event/818757/#9-rhul-trieste
    https://indico.cern.ch/event/853388/#8-prague
  - unfolding becomes *just* a maximum likelihood fit, with one POI per truth-distribution bin
  - systematic uncertainties naturally included in the unfolding procedure, without the need of running toys, build pseudo-data...

- **Why it could be a good idea?**
  - a step forward in the direction of coherent statistical procedure for different types of measurements
  - combination of measurements or channels becomes much easier
  - possibility to constrain systematics and/or simultaneous determination of background normalization in a single fit (if more bins or control regions added)

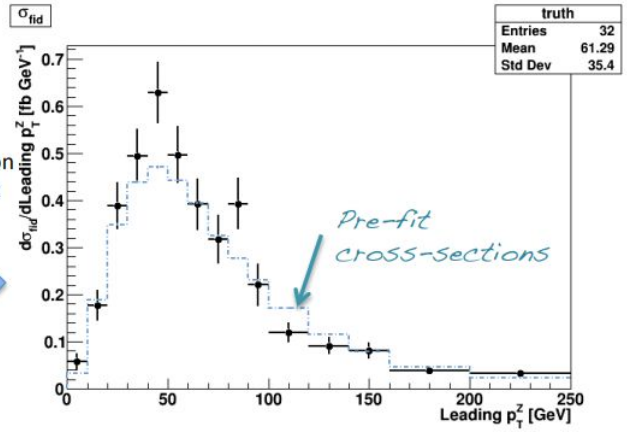
The concept

- Define N “signal” samples (“**sub-samples**” in the following)
  - sample j = distribution of jth truth bin in the reco bins
  - derived using **response matrix** (encoding dependence on systematic effects)
- Perform **simultaneous fit** of reco distribution to the sum of the distributions for the N “sub-samples” ⇒ derive N normalization coefficients, one for each “sub-sample”
  - interpreted as measured cross-section
- Use **profile-likelihood** to perform the fit
  ⇒ use existing techniques to profile systematics, monitor fit, perform sanity checks





Read off cross-section for each bin to plot unfolded result

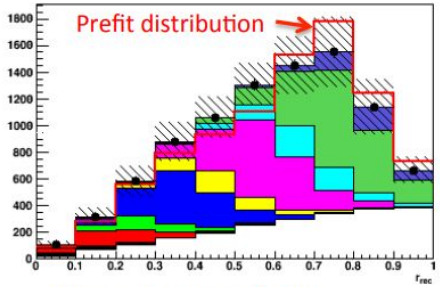


“Does it pass the coffee-break explanation test?”

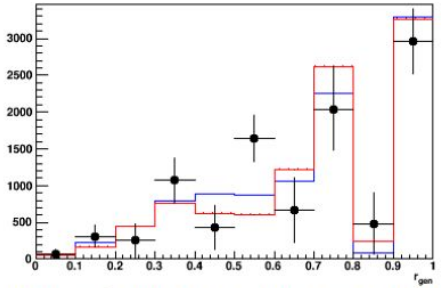
Adding regularization

By W. Buttinger

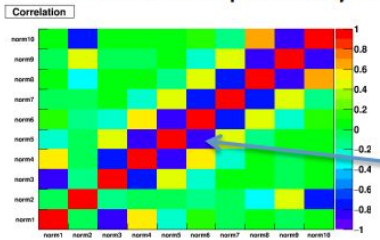
- Apart from profiling systematics, approach of previous slide equivalent to **matrix inversion**
  ⇒ issues with amplification of data fluctuations / anticorrelation between fitted cross-sections



Fit worked perfectly ....

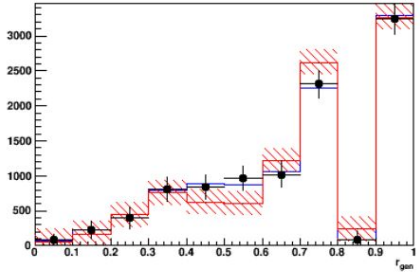


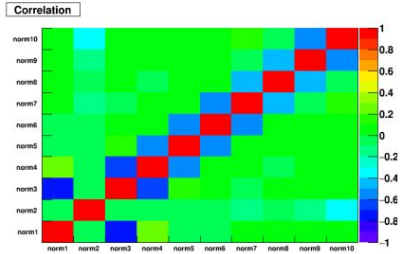
But unfolded result has oscillations about the true distribution (blue) (red is the MC truth distribution)



Strong anti-correlations to neighbour bins

$\tau = 10^{-2.25}$





- Can include **regularization** as additional ***likelihood term***:
  - Entropy- based term
  - Derivative-based term : Tikhonov $e^{-\frac{1}{2}\chi^2} = e^{-\frac{1}{2}\tau^2\Sigma(x_i - x_i^{ref})^2}$

Equivalent to a set of gaussian constraints, one for each bin

Profile likelihood unfolding in TRExFitter

- **TRExFitter** extensively used now by many TopWG analyses
  - even for non-total-cross-section measurements (top mass, top width...)

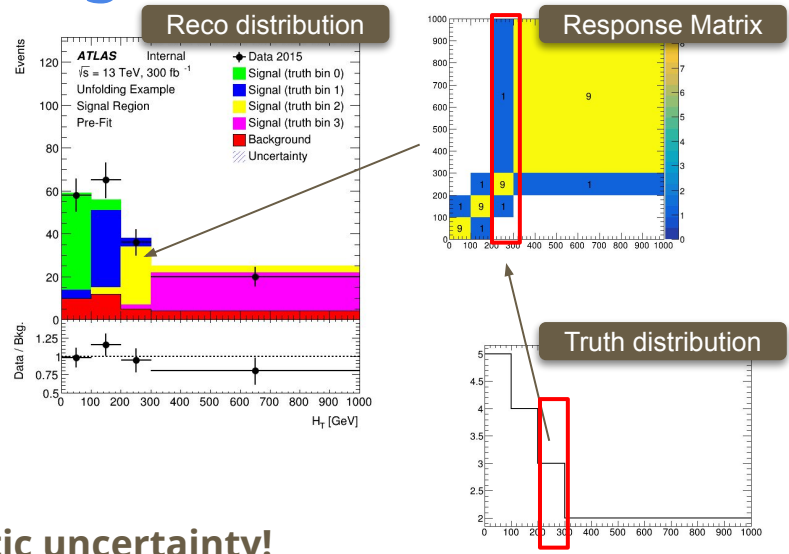
- All the **basic ingredients** for PLU already present in **TRExFitter**
  - basic ***regularization*** added recently to the code
    (*just a matter of adding constraint term to likelihood*)
    - can be easily extended with other regularization methods

- Until few weeks ago, user had to create **inputs for unfolding** by themself, by creating sub-samples as in previous slides
  - recently implemented **specific code** to make this procedure automatic, by giving as inputs **truth distributions** and **response matrices**

- Development documented in JIRA: https://its.cern.ch/jira/browse/TTHFITTER-261
- TRExFitter v4.08 has functionality fully supported:
  https://gitlab.cern.ch/TRExStats/TRExFitter/-/tags/TRExFitter-00-04-08

Implementation - basic idea: Folding

- **"Fold"** each ***truth bin*** of the signal via ***response matrix***
  - get one ***reco distribution*** for each truth bin
  - done **automatically** now
- Assign one **normalisation factor** (*free parameter, if no regularization added*) for normalisation of each "folded truth bin" (i.e. reco signal distribution)
  - effectively one NF per truth bin
- Add **background** reco distributions as usual
- **Systematics** on folded signal:
  - **NB: need a response matrix for each systematic uncertainty!**
- ***Fit*** signal normalisation factors ⇒ ***adjust*** truth bin contents ⇒ get ***unfolded*** distribution
- Possibility to add *Tikhonov tau* parameters for **regularisation**, as additional **constraint terms** in the likelihood for the truth-bin normalization factors
- Can use the full fit machinery that is already available
  - pull plots, correlation matrix, postfit plots, toys, etc...
- Example config: https://gitlab.cern.ch/TRExStats/TRExFitter/blob/master/test/configs/FitExampleUnfolding.config



Implementation - configuration

- New **`FitType`** option in the **`Fit`** block: **`UNFOLDING`**
- **New step**: "**u**" that needs to run before "**h**" - does the folding
- New **`Unfolding`** block
  - some basic options, e.g. response matrix orientation, number of truth bins, Tau, cosmetics
- New **`TruthSample`** block(s)
  - path to truth sample(s), cosmetics allowing multiple truth samples to be plotted
- New **`UnfoldingSample`** block(s)
  - path to samples that should be used for folding, cosmetics
- New **`UnfoldingSystematic`** block(s)
  - path to systematic response matrices, automatically propagates for signal sample

- **Response matrix can be built from**:
  - combination of selection eff, migration matrix and acceptance (for fiducial)
  - or directly read from a file

Configuration - Unfolding block

```
Unfolding: "Unfolding"
  MatrixOrientation: TRUTHONVERTICAL
  TruthDistributionPath: test/inputs/Unfolding
  TruthDistributionFile: InputUnfolding
  TruthDistributionName: nominal/leadingTop_pt_signal_truth
  NumberOfTruthBins: 11
  Tau: 1:2,3:2
  TitleX: "Top p_{T} [GeV]"
  TitleY: "Events"
  RatioYmax: 2.0
  RatioYmin: 0.3
  LogX: FALSE
  LogY: FALSE
  TitleOffsetX: 2.5
  TitleOffsetY: 1.2
  NominalTruthSample: "PP8"
  UnfoldingResultMin: -100
  UnfoldingResultMax: 100
```

Required:

- **NumberOfTruthBins**
- **NominalTruthSample**
  - sets which **TruthSample** is the one that should be used for folding
- **UnfoldingResultMin/Max**
  - sets the limits on the NF

Configuration - TruthSample block

```
TruthSample: "PP8"
  Title: "Powheg+Pythia 8"
  LineColor: 2
  FillColor: 0
  TruthDistributionName: nominal/leadingTop_pt_signal_truth

TruthSample: "PH7"
  Title: "Powheg+Herwig 7"
  LineColor: 3
  FillColor: 0
  TruthDistributionName: PS/leadingTop_pt_signal_truth
```

```
Region: "SR1"
  Type: SIGNAL
  HistoName: "pt"
  VariableTitle: "p_{T} [GeV]"
  Label: "Signal Region1"
  ShortLabel: "SR1"
  NumberOfRecoBins: 11
  MigrationFile: "InputUnfolding"
  SelectionEffFile: "InputUnfolding"
```

- Simple path setup (combines with paths with **Job** and **Region** blocks)
  - as is done for standard ntuples/histograms

Configuration - UnfoldingSample block

```
UnfoldingSample: "SignalSample1"
  Title: "SignalSample1"
  FillColor: 3
  LineColor: 3
  MigrationFile: InputUnfolding
  MigrationName: nominal/leadingTop_pt_migration
  SelectionEffFile: InputUnfolding
  SelectionEffName: nominal/efficiency_selection
  AcceptanceName: nominal/acceptance
```

- Sample that is a result of the folding of the **TruthSample** specified
- Can set selection efficiency, migration matrix and acceptance
- Response = SelEff * Migration / Acceptance
  - SelEff < 1 in each bin, Acceptance < 1 in each bin
- Currently: SelEff, Acceptance is only TH1 - not 100% correct (efficiency has binomial statistics), Migration (response) is TH2 object

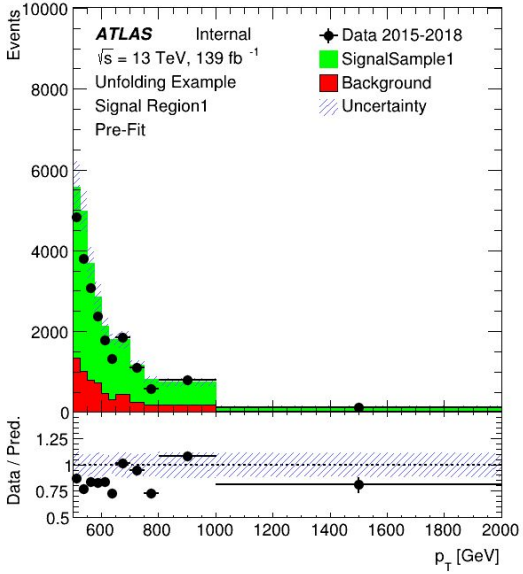
Configuration - UnfoldingSystematic block

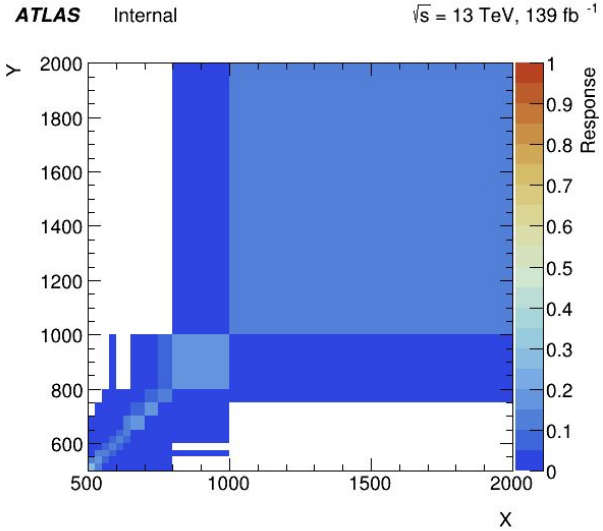
```
UnfoldingSystematic: "bTag_SF_B_eigen0"
  Title: "Btag_B_0"
  NuisanceParameter: "bTag_SF_B_eigen0"
  Type: HISTO
  Samples: "SignalSample1"
  Symmetrisation: TWOSIDED
  MigrationNameDown: "bTagSF_B_eigen0_1down/leadingTop_pt_migration"
  MigrationNameUp: "bTagSF_B_eigen0_1up/leadingTop_pt_migration"

UnfoldingSystematic: "PS"
  Title: "PS"
  Type: HISTO
  Samples: "SignalSample1"
  Symmetrisation: ONESIDED
  MigrationNameUp: "PS/leadingTop_pt_migration"
```

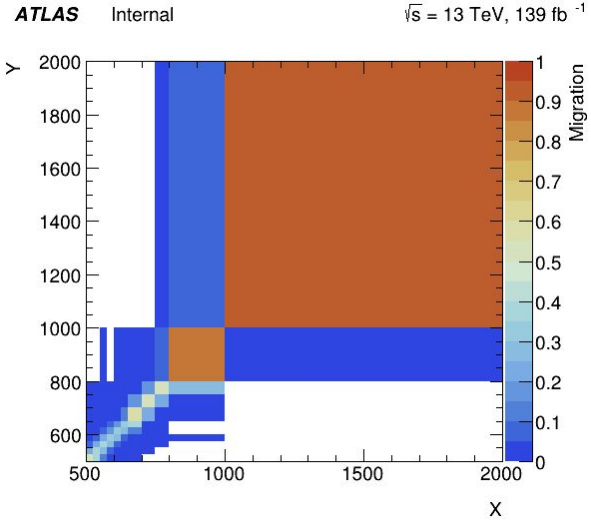
- Support for **ONESIDED** and **TWOSIDED**
- Similar syntax as for **Systematic**

Toy example

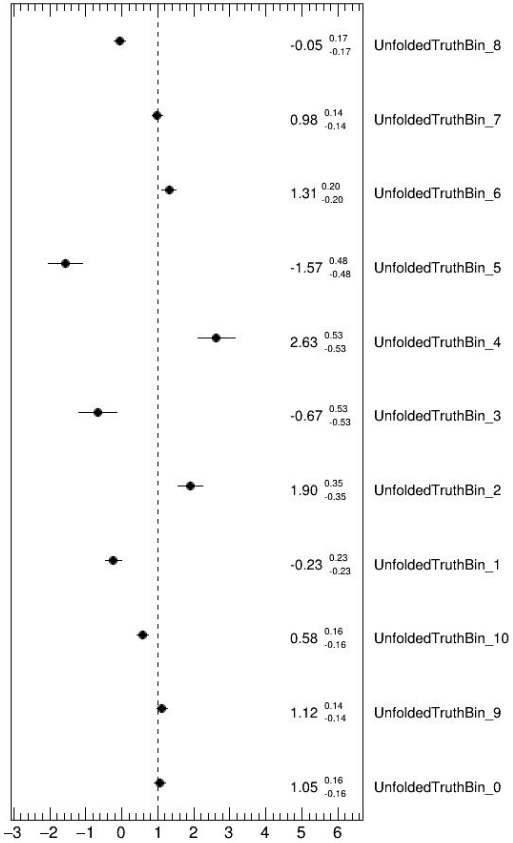
- Inputs given by all-had tt XS analysis (**Jiri Hejbal, Roman Lysak**)
  - slightly rearranged to fully fit the needed input structure
- Using exact same config file as in previous slides and .root file with all input histograms (both included in the repository now)



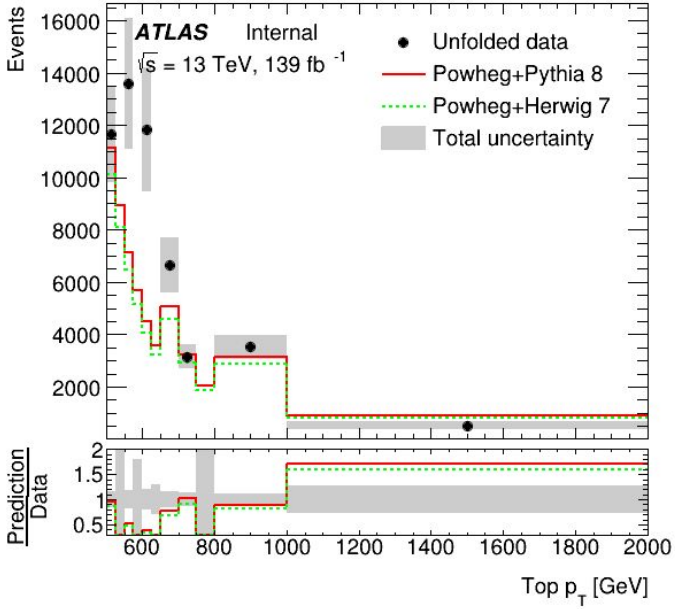


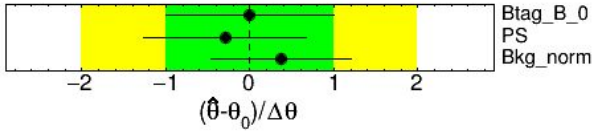


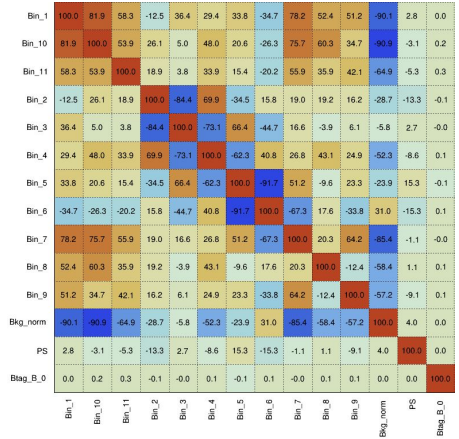
Toy example - Fit results



- Fitted values of NFs used to create unfolded result plot
  - error band = NF uncertainty ⇒ includes all uncertainties
- Usual TRExFitter nuisance parameter pull plot and correlation matrix
  - why nuisance parameter pulls? …

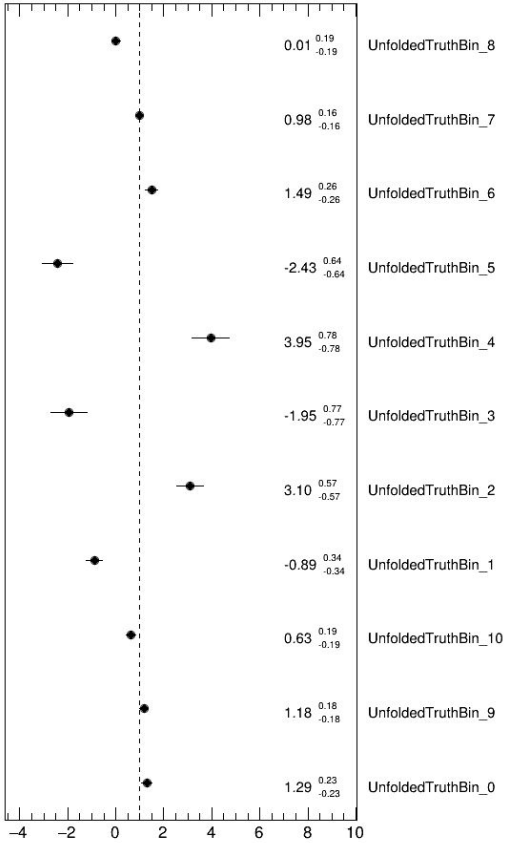




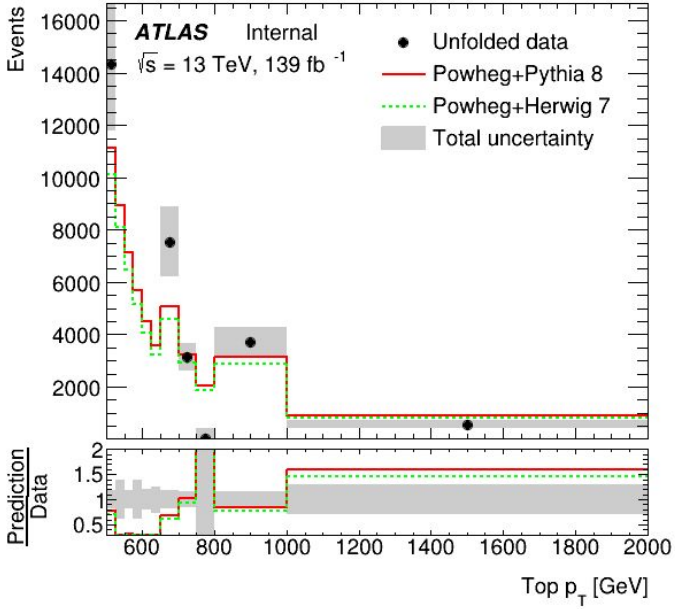


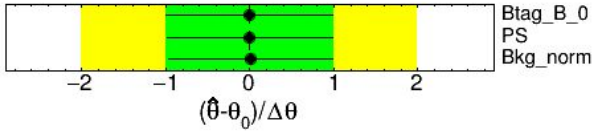
	Bin_1	Bin_10	Bin_11	Bin_2	Bin_3	Bin_4	Bin_5	Bin_6	Bin_7	Bin_8	Bin_9	Bkg_norm	PS	Btag_B_0
Bin_1	100.0	81.9	58.3	-12.5	36.4	29.4	33.8	-34.7	78.2	52.4	51.2	-90.1	2.8	0.0
Bin_10	81.9	100.0	53.9	26.1	5.0	48.0	20.6	-26.3	75.7	60.3	34.7	-90.9	-3.1	0.2
Bin_11	58.3	53.9	100.0	18.9	3.8	33.9	15.4	-20.2	55.9	35.9	42.1	-64.9	-5.3	0.3
Bin_2	-12.5	26.1	18.9	100.0	-84.4	69.9	-34.5	15.8	19.0	19.2	16.2	-28.7	-13.3	-0.1
Bin_3	36.4	5.0	3.8	-84.4	100.0	-73.1	66.4	-44.7	16.6	-3.9	6.1	-5.8	2.7	-0.0
Bin_4	29.4	48.0	33.9	69.9	-73.1	100.0	-62.3	40.8	26.8	43.1	24.9	-52.3	-8.6	0.1
Bin_5	33.8	20.6	15.4	-34.5	66.4	-62.3	100.0	-91.7	51.2	-9.6	23.3	-23.9	15.3	-0.1
Bin_6	-34.7	-26.3	-20.2	15.8	-44.7	40.8	-91.7	100.0	-67.3	17.6	-33.8	31.0	-15.3	0.1
Bin_7	78.2	75.7	55.9	19.0	16.6	26.8	51.2	-67.3	100.0	20.3	64.2	-85.4	-1.1	-0.0
Bin_8	52.4	60.3	35.9	19.2	-3.9	43.1	-9.6	17.6	20.3	100.0	-12.4	-58.4	1.1	0.1
Bin_9	51.2	34.7	42.1	16.2	6.1	24.9	23.3	-33.8	64.2	-12.4	100.0	-57.2	-9.1	0.1
Bkg_norm	-90.1	-90.9	-64.9	-28.7	-5.8	-52.3	-23.9	31.0	-85.4	-58.4	-57.2	100.0	4.0	0.0
PS	2.8	-3.1	-5.3	-13.3	2.7	-8.6	15.3	-15.3	-1.1	1.1	-9.1	4.0	100.0	0.0
Btag_B_0	0.0	0.2	0.3	-0.1	-0.0	0.1	-0.1	0.1	-0.0	0.1	0.1	0.0	0.0	100.0

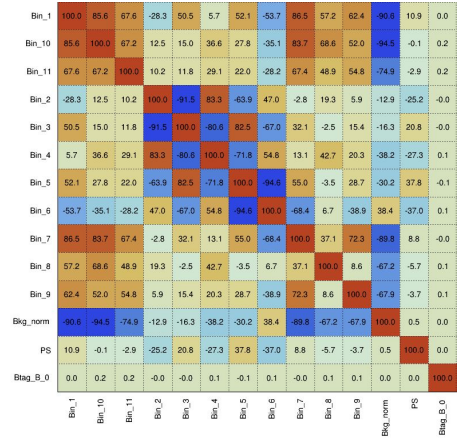
Toy example - Fit results



- We had “tau” set in the config ⇒ regularization was ON for some of the bins
- Switching it OFF:



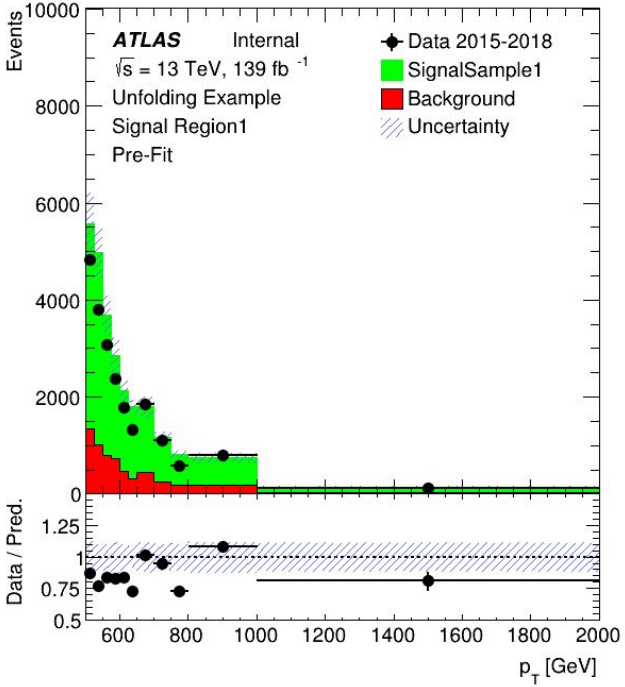




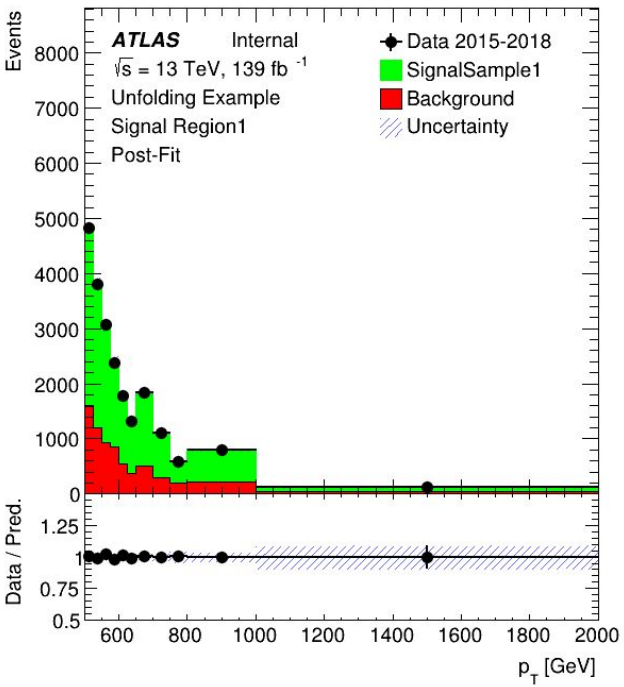
	Bin_1	Bin_10	Bin_11	Bin_2	Bin_3	Bin_4	Bin_5	Bin_6	Bin_7	Bin_8	Bin_9	Bkg_norm	PS	Btag_B_0
Bin_1	100.0	85.6	67.6	-28.3	50.5	5.7	52.1	-53.7	86.5	57.2	62.4	-90.6	10.9	0.0
Bin_10	85.6	100.0	67.2	12.5	15.0	36.6	27.8	-35.1	83.7	68.6	52.0	-94.5	-0.1	0.2
Bin_11	67.6	67.2	100.0	10.2	11.8	29.1	22.0	-28.2	67.4	48.9	54.8	-74.9	-2.9	0.2
Bin_2	-28.3	12.5	10.2	100.0	-91.5	83.3	-63.9	47.0	-2.8	19.3	5.9	-12.9	-25.2	-0.0
Bin_3	50.5	15.0	11.8	-91.5	100.0	-80.6	82.5	-67.0	32.1	-2.5	15.4	-16.3	20.8	-0.0
Bin_4	5.7	36.6	29.1	83.3	-80.6	100.0	-71.8	54.8	13.1	42.7	20.3	-38.2	-27.3	0.1
Bin_5	52.1	27.8	22.0	-63.9	82.5	-71.8	100.0	-94.6	55.0	-3.5	28.7	-30.2	37.8	-0.1
Bin_6	-53.7	-35.1	-28.2	47.0	-67.0	54.8	-94.6	100.0	-68.4	6.7	-38.9	38.4	-37.0	0.1
Bin_7	86.5	83.7	67.4	-2.8	32.1	13.1	55.0	-68.4	100.0	37.1	72.3	-89.8	8.8	-0.0
Bin_8	57.2	68.6	48.9	19.3	-2.5	42.7	-3.5	6.7	37.1	100.0	8.6	-67.2	-5.7	0.1
Bin_9	62.4	52.0	54.8	5.9	15.4	20.3	28.7	-38.9	72.3	8.6	100.0	-67.9	-3.7	0.1
Bkg_norm	-90.6	-94.5	-74.9	-12.9	-16.3	-38.2	-30.2	38.4	-89.8	-67.2	-67.9	100.0	0.5	0.0
PS	10.9	-0.1	-2.9	-25.2	20.8	-27.3	37.8	-37.0	8.8	-5.7	-3.7	0.5	100.0	0.0
Btag_B_0	0.0	0.2	0.2	-0.0	-0.0	0.1	-0.1	0.1	-0.0	0.1	0.1	0.0	0.0	100.0

Toy example - Post-fit plots

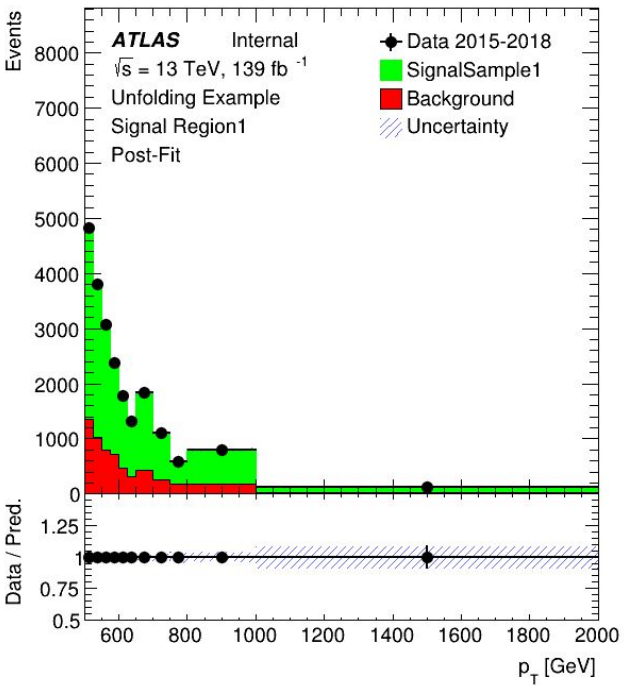
Pre-fit



Post-fit with regularization



Post-fit w/o regularization



Properties of the fit

- **Properties of the unfolding fit**:
  - if **squared migration matrix**, **no regularization** AND norm factor **range large** enough ⇒ **no pulls/constraints**, post-fit plot data/MC agreement perfect by construction (*system of N linear equations with N free parameters*)
  - otherwise, can have pulls and constraints

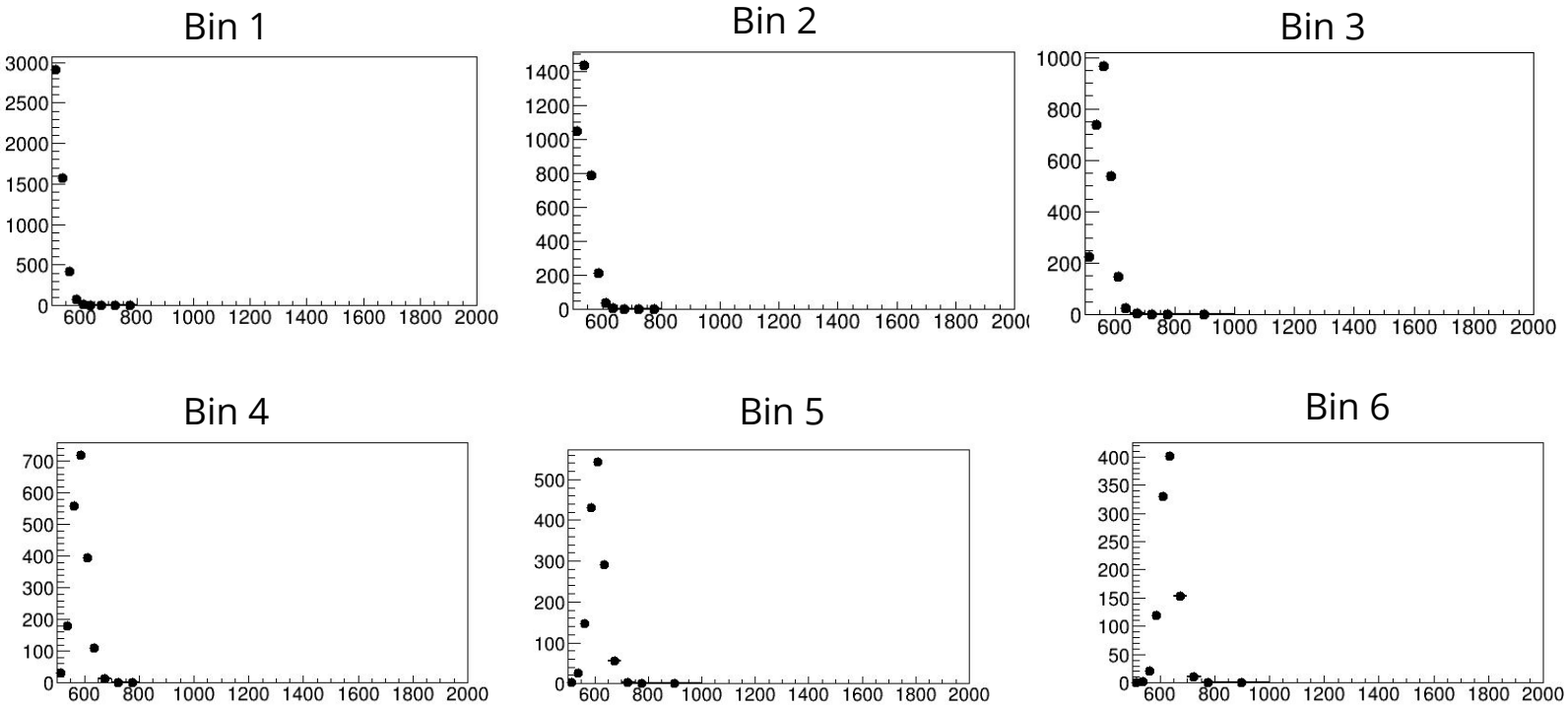
- **Nice features of the method**:
  - can include **free-floating** parameters for backgrounds
  - can include **control regions**
  - can have **more reco bins** than truth bins
  - can **combine** regions/channels to unfold a single distribution (same as FBU)
  - **self-consistent & coherent** approach:
    - same method for fitting and unfolding between analyses
    - no need to use two different techniques in a single paper e.g. for total and differential cross-section

Summary and plans

- First complete implementation of **profile-likelihood unfolding (PLU)** in **TRExFitter**
  - analyses can **start testing** the functionality
  - at the moment fully working only when using **histograms as inputs** (*not with ntuples*)

- **Plans**:
  - include functionalities to help with **stress tests**
  - add possibility to use **ntuples** as inputs (i.e. on-the-fly response matrix creation)
  - **MC stat-error** on signal templates → thinking about a way to get it 100% correct
  - **smoothing** of systematic variations on signal templates
  - add more **regularization** options
  - think about unfolding of **normalized cross-sections**

- **Comparison with other unfolding methods**:
  - **feedback** very welcome in this sense
  - **performance** should be **the same**, in "**simple conditions**" (*i.e. square migration, either with no regularization or for a certain tuned value of Tikhonov regularization term*)

Backup

Folding example - folded truth bins